MILITARY SPECIFICATION

PROJECTILE, 175MM: HE, M437
METAL PARTS ASSEMBLY

This amendment forms a part of Military Specification MIL-P-60622B (AR), dated 14 September 1977 and is approved for use by the U.S. Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 9

4.4.1: Delete in its entirety and substitute:

"4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, heat treatment lots shall contain body forgings from not more than one heat of steel that has been heat treated in the same heat treatment equipment, at the same temperature, in one unchanged process. The heat treatment lot thus constituted shall be the basis for selecting the samples for the tension test."

PAGE 13

4.4.2.4: Add Critical defect 2:

"2 Missing base plate, broken base plate weld or perforated base plate 100% Visual".

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4.4.3.2.2: Delete in its entirety and substitute:

"4.4.3.2.2 Tension test. Two projectile bodies, one of the hardest and one of the softest as determined by 4.4.3.2.1 shall be subjected to this test. Failure of any tension test specimen in the side wall to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the lot. Should a tension test failure occur in the base sample, another sample from another location in the base of the same projectile may be taken. At the option of the supplier, a production control sample may be utilized to

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establish a hardness range. When a production control sample is used, the hardness range thus established shall be maintained throughout the heat treatment lot. The size of the sample shall be determined by the contractor. Projectile bodies which fall outside of the established range may be re-heat treated to fall within the range, or new mechanical properties may be determined from newly selected projectile bodies which will represent the true extremes of hardness in the heat treatment lot."

PAGE 22

* Add the following new paragraph 4.4.3.4:

"4.4.3.4 Salt spray. After qualification in accordance with TT-C-490, two (2) salt spray samples shall be taken each week, one during the first 30 minutes of production and a second at any other time. Failure to comply with the requirements of the applicable drawing shall be cause for taking the corrective action prescribed in TT-C-490."

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* 4.5.1.1: Delete in its entirety and substitute the following:

"4.5.1.1 Salt spray testing. Salt spray testing shall be accomplished on production projectiles or sectioned projectiles that were acceptable at the time of painting and were sectioned subsequent to painting. If sectioned projectiles are used, sectioning shall be accomplished to allow testing of all major areas of the projectiles on an alternating basis. The salt spray test shall be performed utilizing the equipment and procedures specified by Federal Test Method Standard No. 141 and TT-C-490 with the exception that test samples shall be selected in accordance with 4.4.3.4 and test samples shall be dried in accordance with Drawing 10520193."

* 4.5.3.4: Delete in its entirety and substitute the following:

"4.5.3.4 Operator qualifications. For operation of ultrasonics equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level I. For set-up and calibration of ultrasonics equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level II, in all respects but need not include experience."

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4.5.4.1: Delete in its entirety and substitute the following:

"4.5.4.1 Tension test. From the projectiles selected in accordance with 4.4.3.2.2, three tension test specimens will be taken conforming to the largest obtainable "round type" specimen as prescribed by ASTM Method E-8. Two specimens shall be taken longitudinally, from opposite sides of the body in such a manner that the center of the gage length of the machined test bar will fall in line coincident with the center of the rotating band seat. The third specimen shall be taken transversely through the base. The axis of the base specimen shall be .5 + .1 inches from the centerline of the projectile and .6 + .1 inches from the bottom of the base (see Figure 1). The number of specimens and their location are for the convenience of the contractor and Government and shall not restrict the contractor or Government from testing additional specimens at other locations to determine that the mechanical properties of the projectile meet the drawing requirements. The criteria for performing the tests shall be as prescribed by ASTM Method E-8. Yield strength shall be determined by the offset method. The limiting set shall be 0.1 percent.

PAGE 25

4.5.5, Line 3: Delete the word "reject" and subsitute the word "select".

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4.5.7.1: First sentence, delete the following words: "for recovery".

Delete last sentence in its entirety.

The margins of this amendment are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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